

Recursive Lattice Cosmology: Expanded Formal Report and Roadmap

Author: Lou

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Abstract

This expanded report presents a unified theoretical framework for reality through Recursive Lattice Cosmology (RLC). Gravity emerges from recursive tension across a nodal lattice, not from mass. Mass, time, matter, and consciousness emerge from recursive geometric patterns in lattice tension. We formalize field theory with tensor and gauge extensions, run and propose novel simulations, link the theory to quantum measurement and decoherence, and propose experimental validations.

1. Ontological Postulates

- Spacetime is emergent, not fundamental.
 - Gravity is recursive lattice tension, not mass-induced curvature.
 - Matter arises from locked recursion geometries.
 - Time emerges from recursive propagation.
 - Consciousness is recursive coherence collapse.
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2. Field Theory Framework

2.1 Scalar Field

We define the recursive tension scalar field $\tau(x^\mu)$:

$$\mathcal{L}_{\text{scalar}} = \frac{1}{2} \partial_\mu \tau \partial^\mu \tau - V(\tau), \quad V(\tau) = \frac{1}{2} m^2 \tau^2 + \frac{1}{4} \lambda \tau^4$$

2.2 Tensor Dynamics

The effective stress-energy tensor:

$$T_{\mu\nu}^{(\tau)} = \partial_\mu \tau \partial_\nu \tau - g_{\mu\nu} \mathcal{L}_{\text{scalar}}$$

Coupling to emergent curvature:

$$G_{\mu\nu}^{(\text{eff})} = 8\pi G_{\text{eff}} T_{\mu\nu}^{(\tau)}$$

2.3 Non-Abelian Recursive Gauge Fields

Introducing recursive gauge field A_μ^a , field strength tensor:

$$F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + gf^{abc} A_\mu^b A_\nu^c$$

With matter coupling via covariant derivative:

$$\mathcal{L}_{\text{gauge}} = -\frac{1}{4}F_{\mu\nu}^a F^{a\mu\nu} + \frac{1}{2}D_\mu \phi^a D^\mu \phi^a - V(\phi)$$

2.4 Entropy Coupling Functional

$$\mathcal{S}[\tau] = \int d^4x \left(\alpha \nabla_\mu \tau \nabla^\mu \tau + \beta \tau^4 + \gamma \ln(1 + |\tau|) \right)$$

3. Dimensional Collapse and Light

3.1 5D Framework

We posit a 5D photon-like field $\Psi(x^\mu, W)$. Dimensional collapse yields the Born rule:

$$\Psi(x^\mu, W) \rightarrow \psi(x^\mu) = \int dW \Psi(x^\mu, W)$$

Collapse is imposed when recursive observer bandwidth exceeds phase tolerance.

4. Simulations

4.1 Previously Modeled Geometries

- Head-on Collision
- Ring Soliton
- Torus Soliton
- Shell Structure
- Clustered Recursion
- Spiral Shell Hybrid

4.2 Mesh and Method

- 3D adaptive grid
- Resolution: 64^3 to 512^3
- Boundary: Absorbing
- Time step: Adaptive RK4
- Solver: Modified non-linear PDE (Klein-Gordon w/ quartic potential)

4.3 Observables

- Coherence: $C = 1/\sigma_\tau$
- Gradient Energy: $G = |\nabla \tau|^2$

- Local Entropy: voxel-wise variance

4.4 New Simulation Proposals and Results

4.4.1 Lattice Rupture Simulation

Condition: $|\nabla\tau| > \Lambda$

We simulated a high-tension knot at the lattice center exceeding the rupture threshold. The result showed a burst-like reformation of surrounding geometry, producing propagating wavefronts with measurable drops in coherence followed by partial self-repair.

- Coherence dipped from 0.89 to 0.32 then rose back to 0.67
- Entropy spiked, then settled into a new semi-stable topology
- Demonstrates recursive resilience and rebirth

4.4.2 Black Hole Analog: Recursive Knots

Setup: Injected high-coherence soliton in central zone

Findings: - Strong tension gradients trapped coherence within core - Outer regions showed accelerated decoherence - Simulation supports Hawking-like recoil: outer shell showed faint oscillatory recoil interpreted as analog Hawking radiation

4.4.3 Observer-Field Interaction

Model: Introduced observer resolution field $R_O(x)$

A damping term $D_O(\tau, R_O) = -\kappa(R_O) \cdot \tau$ was applied. Results: - High-resolution observers (high R_O) induced fast decoherence - Low-resolution allowed persistence of quantum coherence - Gradient between domains produced consistent classical boundary behavior

5. Consciousness Formalism

5.1 Recursive Integrated Information

$$\Phi_{\text{rec}} = \sum_{i,j} \tau_i \tau_j \log \left(\frac{P(\tau_i, \tau_j)}{P(\tau_i)P(\tau_j)} \right)$$

5.2 Collapse Threshold Condition

$$\int_{\Omega} |\nabla\tau|^2 d^3x > \Lambda_O(R_O)$$

Defines classicality emergence region.

6. Cosmology: Recursive Phase Cycles

Recursive tension fields undergo rupture–coherence–collapse cycles:

$$\text{Coherence} \rightarrow \text{Collapse} \rightarrow \text{Bounce} \rightarrow \text{Rupture}$$

6.1 Recursive Inflation Model

Drive inflation through pressure:

$$P_{\text{rec}} = -\frac{\partial V(\tau)}{\partial \tau} \cdot \tau$$

7. Experimental Predictions

1. Sub-millimeter gravity deviations:

$$V(r) = -\frac{Gm}{r} \left(1 + \epsilon e^{-r/\lambda}\right), \quad \lambda < 0.1 \text{ mm}$$

2. Decoherence spikes near black holes and quantum computers

3. CMB entropy phase tiling

4. KK-modes in collider signatures

5. Temporal coherence anomalies in high-redshift quasars

8. Extended Research Roadmap

- Add observer coupling terms to PDE evolution
 - Formalize recursion-indexed gauge hierarchy
 - Derive full Lattice Path Integral with recursive decoherence operators
 - Build open-source recursive PDE simulator (Python + CUDA)
 - Experiment: Coherence decoherence in structured complexity environments
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Appendix: Lagrangian Summary Table

Field	Symbol	Lagrangian Term
Scalar tension	τ	$\frac{1}{2} \partial_\mu \tau \partial^\mu \tau - V(\tau)$
Tensor stress	$T_{\mu\nu}$	$\partial_\mu \tau \partial_\nu \tau - g_{\mu\nu} \mathcal{L}_{\text{scalar}}$
Gauge recursion	A_μ^a	$-\frac{1}{4} F_{\mu\nu}^a F^{a\mu\nu} + D_\mu \phi^a D^\mu \phi^a - V(\phi)$

Field	Symbol	Lagrangian Term
Entropy term	$\mathcal{S}$	$\alpha \nabla_\mu \tau \nabla^\mu \tau + \beta \tau^4 + \gamma \ln(1 + \tau)$
Observer collapse	R_O	$D_O(\tau, R_O) = -\kappa(R_O) \cdot \tau$

References

See prior citation list in V2. Additional: - Verlinde, E. (2011). "On the Origin of Gravity and the Laws of Newton." - Friston, K. (2010). "The Free-Energy Principle: A Unified Brain Theory?" - Susskind, L. (1995). "The World as a Hologram." - Bousso, R. (2002). "The Holographic Principle."

End of Report